

Ubuntu va OS asoslari: ma'lumotnoma

Atamalar va commandlar jadvali

Ubuntu Server 24.04 LTS / ochiq Lab v1.0.0
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Qamrov: Ubuntu Server 24.04 LTS / ochiq Lab v1.0.0. Bu yodlash uchun jadval emas. Maqsadga mos commandni toping, kerakli option va tekshirish usulini kitobdan qayta o'qing.

Atamalar

Atama	O'zbekcha izoh
OS (operating system)	Apparat qismlarini boshqarib, process, file, kirish nazorati va aloqa kabi umumiy imkoniyatlarni beradigan asosiy tizim.
kernel	Ishlayotgan OSning markazi; CPU, xotira, qurilma, process, file va tarmoqni boshqaradi.
user space	shell, command va service kabi odatdagi programlar kerneldan ajratilgan holda ishlaydigan makon.
distribution	Linux kerneli, vositalar, packagelar, boshlang'ich sozlamalar va yangilash siyosati birlashtirilgan tizim. Ubuntu bunga misol.
CLI (command-line interface)	Ko'rsatma va natija matn orqali almashiladigan boshqarish usuli.
terminal	Matn kiritish va natijani ko'rsatish oynasi yoki interaktiv ulanish muhiti.
shell	Kiritilgan matnni talqin qilib commandni bajaradigan program. Bu muhitda asosan Bash ishlatiladi.
file	Mazmuni va metadatasi bo'lgan asosiy saqlash birligi.
directory	file va boshqa directory nomlarini obyektlarga bog'lab, iyerarxiya hosil qiladigan maxsus file.
path	directorylar iyerarxiyasidan o'tib obyekt manzilini bildiradigan matn.
owner	file yoki directory metadatasida egasi sifatida yozilgan user.
primary group	Har bir userga bitta asosiy a'zolik sifatida birlashtiriladigan group.
supplementary group	Umumiy resurslardan foydalanish kabi maqsadlar uchun qo'shimcha a'zolik.
permission	owner, group va others uchun o'qish (r), yozish (w), bajarish (x) huquqlari.
program	Saqlash qurilmasidagi bajariladigan ko'rsatmalar va tegishli filelar.
process	Xotiraga yuklanib bajarilayotgan program; o'z PIDi va bajarish huquqlariga ega.
package	Dastur filelari, versiya, bog'liqlik va o'rnatish tartibini jamlaydigan tarqatish birligi.
service	Tizim boshqaradigan, fonda davomli imkoniyat beradigan program va uning boshqaruv birligi.
unit	systemd service va boshqa resurslarni boshqaradigan sozlama birligi.
socket	process kernelning tarmoq aloqasi imkoniyatidan foydalanadigan kirish-chiqish nuqtasi.
listen	socket clientning ulanish requestini qabul qilishga tayyor holat.
loopback	Aloqani o'sha host ichiga qaytaradigan virtual tarmoq interfacei; IPv4da 127.0.0.1 misol.
port	Bir host ichidagi turli aloqa nuqtalarini farqlaydigan raqam.
log	program yoki OS ishi, xatosi va holat o'zgarishi haqidagi yozuv.
local	SSH/SCP ulanishi yoki uzatishini boshlaydigan muhit; bu mashqda AWS CloudShell.
remote	SSH/SCP ulanishi yoki uzatilishi manzili; bu mashqda Ubuntu EC2.

Atama	O'zbekcha izoh
authentication	Ulanayotgan tomon da'vo qilgan hisobning haqiqiy egasi ekanini tekshirish.
authorization	Tekshirilgan tomon muayyan resurs bilan nima qila olishini hal qilish.

Bilish zarur bo'lgan commandlar

Maqsad	Command va kerakli options	Xavf yoki noto'g'ri talqin	Keyingi tekshiruv
Hozirgi joy	<code>pwd</code>	Faqat <code>prompt</code> matniga qarab joyni taxmin qilmang.	Chiqqan <code>path</code> ni <code>hostname</code> va <code>user</code> bilan solishtiring.
Joyini o'zgartirish	<code>cd PATH</code> , <code>cd ..</code> , <code>cd ~</code>	<code>relative path</code> joriy <code>directory</code> ga bog'liq.	<code>pwd</code>
Ro'yxat	<code>ls -la</code> , <code>ls -ld DIR</code>	<code>-l</code> <code>DIR</code> ichidagi obyektlarni, <code>-ld</code> <code>DIR</code> <code>directory</code> ning o'zini ko'rsatadi.	Kerak bo'lsa <code>stat</code> yoki <code>tree</code> .
Iyerarxiya	<code>tree PATH</code>	Tuzilishni ko'rsatadi, lekin mazmun va <code>permission</code> ni to'liq bermaydi.	<code>ls -l</code> bilan birga ishlatish.
Directory yaratish	<code>mkdir -p PATH</code>	Noto'g'ri iyerarxiyani yaratib qo'ymang.	<code>tree</code> , <code>ls -ld</code>
Nusxa olish	<code>cp SOURCE DEST</code>	Manba va manzil tartibini adashtirmang; eski <code>file</code> ustidan yozib qo'yishingiz mumkin.	<code>cat</code> , <code>sha256sum</code>
Ko'chirish yoki nomlash	<code>mv SOURCE DEST</code>	Manzildagi mavjud <code>file</code> ustidan yozish xavfi bor.	<code>ls</code>
O'chirish	<code>rm FILE</code>	GUI "savatchasi" yo'q. Bu mashqda <code>rm -rf</code> ni ishlatmang.	Oldin <code>pwd</code> va <code>ls</code> bilan maqsadni, keyin uning yo'qligini tekshiring.
Tahrirlash	<code>nano FILE</code>	To'g'ri <code>fileni</code> saqlaganingizni va saqlamasdan chiqib ketmaganingizni tekshiring.	<code>Ctrl+O</code> , <code>Enter</code> , <code>Ctrl+X</code> ; keyin <code>cat</code> .
Mazmunni ko'rish	<code>cat FILE</code> , <code>less FILE</code>	Ikkilik yoki juda katta <code>fileni</code> ehtiyotsiz <code>cat</code> qilmang.	<code>less</code> dan <code>q</code> bilan chiqing.
Qidirish	<code>grep [-F] PATTERN FILE</code>	<code>-n</code> satr raqamini qo'shadi; talab qilinmasa ishlatmang.	Mos satrlar va <code>exit</code> statusni ko'ring.
Oxirgi satrlar	<code>tail -n N FILE</code>	Bu <code>-n</code> satrlar sonini bildiradi; <code>grep</code> -ndan farq qiladi.	Satrlar soni va tartibini tekshiring.
Chiqishni saqlash	<code>></code> , <code>>></code>	<code>></code> eskisini bo'shatib, ustidan yozadi. Kirish <code>filei</code> bilan bir xil manzilni bermang.	<code>cat</code> , <code>wc -l</code>
Commandlarni ulash	<code>\ </code>	Faqat <code>stdout</code> uzatiladi; <code>stderr</code> odatda o'tmaydi.	Ulamasdan oldin har bir commandni alohida sinang.
User/group	<code>id [USER]</code> , <code>groups [USER]</code>	Ro'yxatdagi hisob bilan ochiq <code>session</code> huquqlarini farqlang.	Yangi <code>session</code> da <code>id</code> bajaring.
Account ma'lumoti	<code>getent passwd USER</code> , <code>getent group GROUP</code>	<code>group</code> ro'yxatida uni <code>primary group</code> qilgan <code>user</code> ko'rinmasligi mumkin.	<code>id</code> <code>USER</code> bilan tekshiring.
Usrni almashtirish	<code>sudo su - USER</code> , <code>exit</code>	<code>shell</code> lar ichma-ich ochiladi; mashq hisoblarida <code>sudo</code> huquqi yo'q.	Har safar <code>id -un</code> va <code>pwd</code> .
Bitta commandni boshqa user sifatida bajarish	<code>sudo -u USER -- COMMAND</code>	<code>root</code> uchun muvaffaqiyat maqsad <code>user</code> uchun ham ruxsat borligini isbotlamaydi.	Darhol <code>echo \$?</code> .
Groupga qo'shish	<code>usermod -aG GROUP USER</code>	<code>-asiz -G</code> eski <code>supplementary</code> <code>grouplardan</code> chiqarishi mumkin.	<code>id</code> <code>USER</code> va yangi <code>session</code> .
Groupdan chiqarish	<code>gpasswd -d USER GROUP</code>	<code>user</code> va <code>group</code> argumentlari tartibini adashtirmang.	<code>id</code> <code>USER</code>
Egasi/groupni o'zgartirish	<code>chown OWNER:GROUP PATH</code>	<code>-R</code> bilan kutilmagan obyektlar ham o'zgarishi mumkin.	<code>stat -c '%U:%G %a %n'</code>
Modeni o'zgartirish	<code>chmod MODE PATH</code>	O'ylamasdan <code>777</code> qo'ymang; <code>x</code> <code>file</code> va <code>directory</code> da turlicha.	<code>stat</code> va boshqa <code>user</code> nomidan <code>test</code> .

Maqsad	<code>command</code> va kerakli <code>options</code>	Xavf yoki noto'g'ri talqin	Keyingi tekshiruv
Processlar ro'yxati	<code>ps -e -o ..., ps -p PID -o ...</code>	Kitobdagi PID misolini aynan ko'chirmang.	<code>user</code> , <code>command</code> nomi va PIDni solishtiring.
Signal yuborish	<code>kill -TERM PID</code>	PID o'zgaruvchisi bo'sh, 0, manfiy yoki 1 emasligini tekshiring. PID uchun <code>pkill</code> ishlatmang.	<code>ps</code> yoki <code>service</code> holatida to'xtaganini ko'ring.
Package ma'lumoti va o'rnatish	<code>apt show NAME</code> , <code>sudo apt update</code> , <code>sudo apt install NAME</code>	<code>apt update</code> ro'yxatni, <code>apt upgrade</code> o'rnatilgan <code>packagelarni</code> yangilaydi. Mashq shartidan tashqari butun tizimni o'zgartirmang.	<code>command -v</code> , <code>dpkg-query</code>
Service sozlamasi	<code>systemctl cat UNIT</code>	Mashq unit fileini bevosita tahrirlamang.	User, WorkingDirectory, ExecStartni o'qing.
Service holati	<code>systemctl status/start/stop</code>	<code>active</code> bo'lish barcha funksiyalar ishlashini isbotlamaydi.	<code>is-active</code> , MainPID, <code>curl</code> bilan tekshiring.
Avtomatik boshlanish	<code>systemctl enable/disable UNIT</code>	Hozir boshlash (start/stop) va keyingi yuklanish (enable/disable) boshqa amallar.	<code>is-enabled</code>
Main PID	<code>systemctl show -p MainPID --value UNIT</code>	PID 0, bo'sh yoki avvalgi ishlashdan qolgan qiymat emasligini tekshiring.	<code>ps -p PID</code>
Socket holati	<code>sudo ss -ltp</code>	Manzil, port va process PIDini birgalikda tekshiring.	<code>service</code> Main PIDi bilan solishtiring.
HTTP aloqa	<code>curl -i URL</code> , zarur bo'lsa <code>--max-time</code>	TCP ulanishi, HTTP status code va response body mazmuni alohida tekshiruvlar.	status line, mazmun va exit status.
Journal log	<code>sudo journalctl -u UNIT --no-pager -n N</code>	Eski ishga tushish loglarini hozirgi log bilan adashtirmang.	Kerakli request path yozilganini toping.
SSH ulanishi	<code>ssh HOST</code> , <code>ssh HOST 'COMMAND'</code>	SSH host alias va DNS hostnameni, local/remote o'zgaruvchi kengayishini, private key himoyasini farqlang.	Ulangach <code>id -un</code> , <code>hostname</code> , <code>pwd</code> .
SCP file uzatish	<code>scp SOURCE DEST</code>	Manba va manzilni almashtirmang; mavjud file ustidan yozish xavfi bor.	Ikki muhitda <code>sha256sum</code> , <code>remoteda</code> stat.

Darsda ishlatiladi, lekin yodlash shart emas

`uname -r`, `hostname`, `find`, `touch`, `stat`, `namei -l`, `test -r/-w`, `echo $?`, `printf`, `command -v`, `dpkg-query`, `dpkg -S`, `sha256sum`, `readlink`, `tr`, `/proc/PID/`.

Maqsad va natijani o'qishni kitobdan tekshiring. Zarur bo'lsa `man`, `help` yoki shu jadvalga qayting.

Keyingi bo'limlarda ko'riladigan commandlar

`ip`, `ping`, `dig`, `tcpdump`, `mount`, `df`, `psql`. Bu kitob faqat aloqasini ko'rsatadi; batafsil sozlash tarmoq, saqlash, DNS va ma'lumotlar bazasi bo'limlarida o'rganiladi.

Bu kitobga kirmaydigan mavzular

kernelni yaratish va `debug` qilish; CPU ko'rsatmalarining ichki bajarilishi; xotira sahifalarini almashtirish algoritmi; disk I/O rejalashtiruvchisini boshqarish; `file system` yaratish yoki bo'limlarni o'zgartirish; tashqi host portlarini `scan` qilish; SSH `server` sozlamasini o'zgartirish.